

DAMM v2

Release 0.2.3

Smart Contract Security Assessment

August 2026

Prepared for:

Meteora

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1 About Offside Labs

Offside Labs is a leading security research team, composed of top talented hackers from both academia and industry.

We possess a wide range of expertise in modern software systems, including, but not limited to, *browsers, operating systems, IoT devices, and hypervisors*. We are also at the forefront of innovative areas like *cryptocurrencies* and *blockchain technologies*. Among our notable accomplishments are remote jailbreaks of devices such as the **iPhone** and **PlayStation 4**, and addressing critical vulnerabilities in the **Tron Network**.

Our team actively engages with and contributes to the security community. Having won and also co-organized *DEFCON CTF*, the most famous CTF competition in the Web2 era, we also triumphed in the **Paradigm CTF 2023** within the Web3 space. In addition, our efforts in responsibly disclosing numerous vulnerabilities to leading tech companies, such as *Apple, Google, and Microsoft*, have protected digital assets valued at over **\$300 million**.

In the transition towards Web3, Offside Labs has achieved remarkable success. We have earned over **\$9 million** in bug bounties, and **three** of our innovative techniques were recognized among the **top 10 blockchain hacking techniques of 2022** by the Web3 security community.



<https://offside.io/>



<https://github.com/offsidelabs>



https://twitter.com/offside_labs



2 Executive Summary

Introduction

Offside Labs completed a security audit of *DAMM v2* smart contracts, starting on August 13th, 2026, and concluding on August 14th, 2026.

Project Overview

Meteora DAMM v2 Release 0.2.3 disables the creation of new pools using the rate limiter fee mode while preserving support for existing pools. It removes two deprecated protocol fee collection paths in favor of the protocol fee wrapper and updates compounding pool accounting so fees attributable to dead liquidity are distributed between protocol fees and pool reserves according to their claiming and compounding components.

Audit Scope

The assessment scope contains mainly the smart contracts of the cp-amm program for the *DAMM v2* project.

The audit is based on the following specific branches and commit hashes of the codebase repositories:

- DAMM v2
 - Codebase: <https://github.com/MeteoraAg/damm-v2>
 - Release 0.2.3: [PR-219](#)
 - Commit Hash: 7ea2e1de339fb8b2d0130304c0f7fd5e487d42ba

The issues described in this report have been resolved by the following commit:

- DAMM v2
 - Codebase: <https://github.com/MeteoraAg/damm-v2>
 - Commit Hash: 7ea2e1de339fb8b2d0130304c0f7fd5e487d42ba

We listed the files we have audited below:

- DAMM v2 Release 0.2.3
 - programs/cp-amm/src/constants.rs
 - programs/cp-amm/src/error.rs
 - programs/cp-amm/src/event.rs
 - programs/cp-amm/src/instructions/initialize_pool/ix_initialize_pool.rs
 - programs/cp-amm/src/instructions/operator/mod.rs
 - programs/cp-amm/src/lib.rs
 - programs/cp-amm/src/params/fee_parameters.rs
 - programs/cp-amm/src/state/fee.rs
 - programs/cp-amm/src/state/operator.rs
 - programs/cp-amm/src/state/pool.rs



Findings

The security audit revealed:

- 0 critical issue
- 0 high issue
- 0 medium issue
- 0 low issue
- 2 informational issues

Further details, including the nature of these issues and recommendations for their remediation, are detailed in the subsequent sections of this report.



3 Summary of Findings

ID	Title	Severity	Status
01	Dead Liquidity Fee Distribution Was Unclear	Informational	Fixed
02	Outdated DAMM v2 IDLs in External Integrations	Informational	Fixed



4 Key Findings and Recommendations

4.1 Informational and Undetermined Issues

Dead Liquidity Fee Distribution Was Unclear

Severity: Informational

Status: Fixed

Target: Smart Contract

Category: Code QA

The documentation did not previously clarify that the dead-liquidity share of claiming fees is allocated to protocol fees, while its share of compounding fees is added back to pool reserves. It is recommended to document these two components separately to make the resulting fee distribution clear.

Outdated DAMM v2 IDLs in External Integrations

Severity: Informational

Status: Fixed

Target: Smart Contract

Category: Code QA

The DBC, Zap, and protocol-fee repositories still contain DAMM v2 IDLs generated from older program versions. Since this release removes two instructions and updates the DAMM v2 IDL, it is recommended to update the bundled IDLs when the respective integrations are next maintained or released.



5 Disclaimer

This audit report is provided for informational purposes only and is not intended to be used as investment advice. While we strive to thoroughly review and analyze the smart contracts in question, we must clarify that our services do not encompass an exhaustive security examination. Our audit aims to identify potential security vulnerabilities to the best of our ability, but it does not serve as a guarantee that the smart contracts are completely free from security risks.

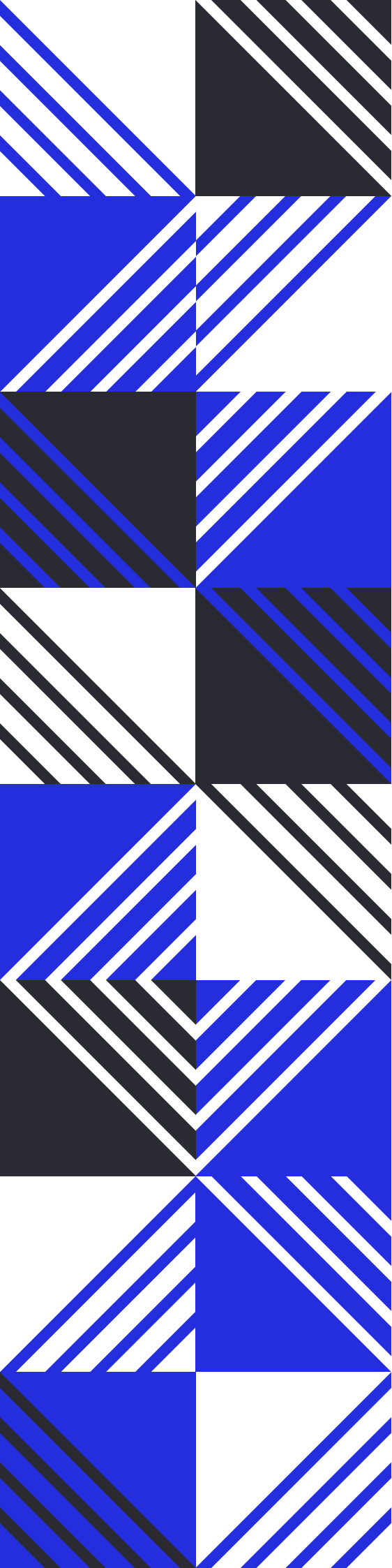
We expressly disclaim any liability for any losses or damages arising from the use of this report or from any security breaches that may occur in the future. We also recommend that our clients engage in multiple independent audits and establish a public bug bounty program as additional measures to bolster the security of their smart contracts.

It is important to note that the scope of our audit is limited to the areas outlined within our engagement and does not include every possible risk or vulnerability. Continuous security practices, including regular audits and monitoring, are essential for maintaining the security of smart contracts over time.

Please note: we are not liable for any security issues stemming from developer errors or misconfigurations at the time of contract deployment; we do not assume responsibility for any centralized governance risks within the project; we are not accountable for any impact on the project's security or availability due to significant damage to the underlying blockchain infrastructure.

By using this report, the client acknowledges the inherent limitations of the audit process and agrees that our firm shall not be held liable for any incidents that may occur subsequent to our engagement.

This report is considered null and void if the report (or any portion thereof) is altered in any manner.



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